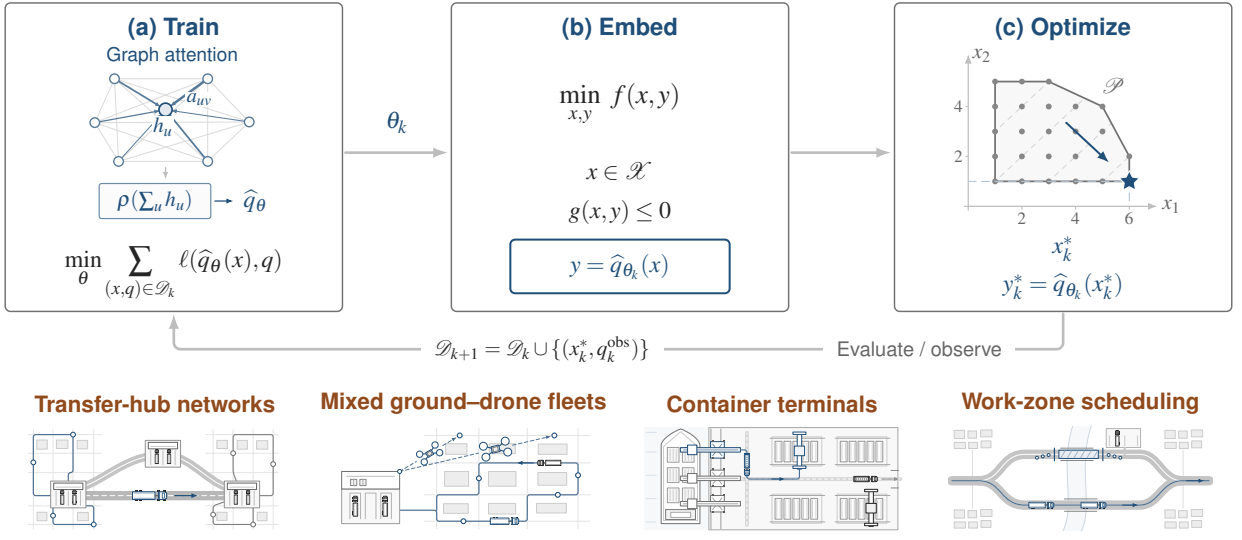

(a) Train
Graph attention
a_{uv}
h_u
$\rho(\sum_u h_u) \rightarrow \widehat{q}_\theta$
$\min_\theta \sum_{(x,q)\in\mathscr{D}_k} \ell(\widehat{q}_\theta(x), q)$
θ_k
(b) Embed
$\min_{x,y} f(x,y)$
$x \in \mathscr{X}$
$g(x,y) \le 0$
$y = \widehat{q}_{\theta_k}(x)$
(c) Optimize
x_2
x_1
$\mathscr{P}$
x_k^*
$y_k^* = \widehat{q}_{\theta_k}(x_k^*)$
$\mathscr{D}_{k+1} = \mathscr{D}_k \cup \{(x_k^*, q_k^{\text{obs}})\}$
Evaluate / observe
Transfer-hub networks
Mixed ground–drone fleets
Container terminals
Work-zone scheduling